1. **Interpret a mathematical result back into the real world context (1)**

   - **Analysis**: This problem requires students to perform calculations involving storage space (350 MB) and then relate the result (remaining space) back to a practical need, judging whether it is "enough" space to store the album. This directly demonstrates the ability to translate a pure mathematical result into a practical real-world conclusion of "can" or "cannot".

2. **Construct and communicate explanations and arguments in the context of the problem (7)**

   - **Analysis**: The question explicitly asks to "show calculations to support your answer". This goes beyond just providing a final "Yes" or "No" answer, requiring students to construct a clear, logical argument supported by calculations and communicate that process.

3. **Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem (2)**

   - **Analysis**: Students need to evaluate whether their calculation results (e.g., available space after deleting certain albums) are sufficient to meet Ivan's need to transfer the 350 MB album. This process involves judging whether the outcome is reasonable or meaningful within the specific context of the problem.

4. **Understand the relationship between the context of the problem and representation of the mathematical solution. Use this understanding to help interpret the solution in context and gauge the feasibility and possible limitations of the solution (9)**

   - **Analysis**: The problem requires students to understand how real-world constraints (e.g., only being able to delete "at most two" music albums) impact the formulation and representation of the mathematical solution. Students must select combinations that adhere to these limits and evaluate feasibility within the real-world context.

5. **Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure or model in order to make contextual judgments about how the results should be adjusted or applied (6)**

   - **Analysis**: The real-world rule ("at most two deletions") directly influences the calculation process and outcomes. Students use this contextual knowledge to determine the optimal calculation method (e.g., choosing to delete the two largest albums to maximize space) to make judgments consistent with the practical situation.

6. **Identify the limits of the model used to solve a problem (3)**

   - **Analysis**: The implicit assumption in the problem is that file sizes are fixed numerical values, but in the real world, storage devices and file management can involve more complex factors. While the question does not directly ask to identify these limits, the problem scenario is built upon a simplified model, and solving it touches upon the boundaries of that model (e.g., precise MB calculation, ignoring file system overhead, etc.).

7. **Recognise (demonstrate, interpret, explain) the extent and limits of mathematical concepts and mathematical solutions (8)**

   - **Analysis**: Similar to point 3, solving this problem requires students to recognize the scope and limitations of basic arithmetic operations when applied to practical storage problems. It demonstrates how mathematical concepts effectively solve these specific, well-structured real-world scenarios.

8. **Use mathematical tools or computer simulations to ascertain the reasonableness of a mathematical solution and any limits and constraints on that solution, given the context of the problem (4)**

   - **Analysis**: Students might use tools like a calculator to aid their calculations (e.g., determining total space after deleting specific albums). While complex simulations are not required, the ability to use tools to verify calculation results is related to this aspect.

9. **Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations in relation to a situation (5)**

   - **Analysis**: The problem provides two formats of information: a pie chart and a table. Students need to extract relevant data from both formats (e.g., free space from the chart, album sizes from the table) and integrate them into the problem-solving process.

1. **Interpret a mathematical result back into the real world context (1)**

   - **分析**: 该问题要求学生进行存储空间大小（350 MB）的计算后，将计算结果（剩余空间）与实际需求联系起来，判断是否“足够”存储相册 (p. 1)。这直接要求将纯数学计算的结果转化为实际的“能”或“不能”的结论，体现了数学结果在现实世界背景下的解释能力。

2. **Construct and communicate explanations and arguments in the context of the problem (7)**

   - **分析**: 题目明确要求“展示计算以支持您的答案” (p. 1)。这不仅仅是提供一个“是”或“否”的最终答案，而是要求学生构建一个逻辑清晰、有计算支持的论证过程，并将此过程传达出来。

3. **Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem (2)**

   - **分析**: 学生需要评估他们的计算结果（例如，删除某些专辑后获得的可用空间）是否足以满足 Ivan 传输 350 MB 相册的需求 (p. 1)。这个过程涉及判断结果在问题的特定情境中是否合理或有意义。

4. **Understand the relationship between the context of the problem and representation of the mathematical solution. Use this understanding to help interpret the solution in context and gauge the feasibility and possible limitations of the solution (9)**

   - **分析**: 这个问题要求学生理解现实限制（例如，只能删除“至多两个”音乐专辑）如何影响数学解决方案的制定和表示 (p. 1)。学生必须选择符合这些限制的删除组合，并在现实背景下评估可行性。

5. **Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure or model in order to make contextual judgments about how the results should be adjusted or applied (6)**

   - **分析**: 现实世界的规则（“至多删除两个”）直接影响了计算过程和结果 (p. 1)。学生需要利用这种背景知识来确定应采用的最佳计算方法（例如，选择删除最大的两个专辑以最大化空间），以做出符合实际情况的判断。

6. **Identify the limits of the model used to solve a problem (3)**

   - **分析**: 问题中隐含的假设是文件大小是固定的数值，但在现实世界中，存储设备和文件管理可能涉及更复杂的因素 (p. 1)。虽然本题没有直接要求识别这些限制，但问题情境是建立在一个简化的模型上的，解决问题的过程间接触及了模型的边界（如精确的MB计算，忽略了文件系统开销等）。

7. **Recognise (demonstrate, interpret, explain) the extent and limits of mathematical concepts and mathematical solutions (8)**

   - **分析**: 与第3点类似，解决此问题需要学生认识到基本算术运算在解决实际存储问题时的适用范围和局限性。它展示了数学概念如何有效地解决这类特定、结构良好的现实问题。

8. **Use mathematical tools or computer simulations to ascertain the reasonableness of a mathematical solution and any limits and constraints on that solution, given the context of the problem (4)**

   - **分析**: 学生可以使用计算器等工具来辅助计算（例如，确定删除特定专辑后的总空间）。虽然不需要复杂的模拟，但使用工具验证计算结果的能力与此方面相关联。

9. **Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations in relation to a situation (5)**

   - **分析**: 问题提供了饼图和表格两种信息格式 (p. 1)。学生需要从这两种格式中提取相关数据（例如，从饼图获取可用空间，从表格获取专辑大小），并将其整合到解决问题的过程中。